Lab Report
Summer 2024

Course Code: **EEE203L**
Course Title: **Electronic Circuits II Laboratory**

Lab Section:**02**

Experiment No.: 5

Name of the experiment: Familiarization with Passive Filters

Date of Submission:

Group No: **3**

Group Member Details:

1. *Name:* **Souvik Barman Ratul**
   I.D. **24121205**
2. *Name:* **Oshiul Alim Ornob**
   I.D. **24121268**
3. *Name:* **Mahdi Zaman**
   I.D. **24121307**
4. *Name:* **Alif Tamjid**
   I.D. **24121308**

Submitted By :

Name : Alif Tamjid

I.D. : 24121308

Submitted To :

Saad Mahbub Chowdhury

Objective:

The purpose of this experiment is to study passive filters (specially low pass and high pass filters). and understand how they allow certain frequencies to pass while blocking others. By constructing RC filter circuits, measuring their output at different frequencies and comparing the results with theoritical cutoff values, we will verify the frequency-selective behaviour of passive filters.

Theoritical Background:

A filter is a circuit that passes some frequency components of a signal and attenuates others. In passive filters, only resistors (R), capacitors (C) and inductors (L) are used.

Two main types experiment are visible.

→ Low pass filter (LPF)
→ High pass filter (HPF)

For an RC circuit, the cutoff frequency is,

$$f_c = \frac{1}{2\pi RC}$$

At f_c, the output voltage drops to 0.707 of it's maximum values (or the power drops to half).

In AC Analysis,

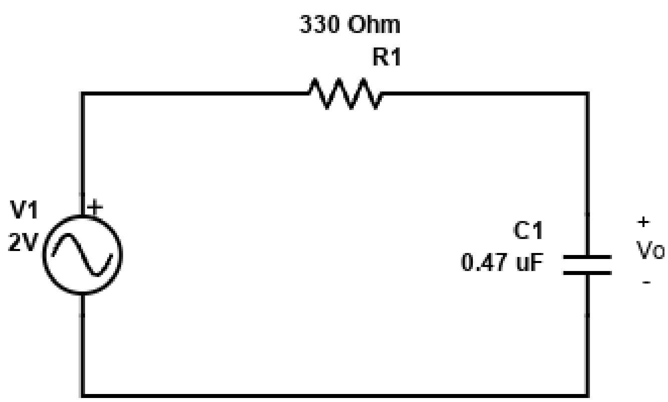
→ At low frequencies, a capacitor offers high reactance $X_c = \frac{1}{2\pi f C}$.

→ At high frequencies, X_c becomes small, allowing more current to pass.

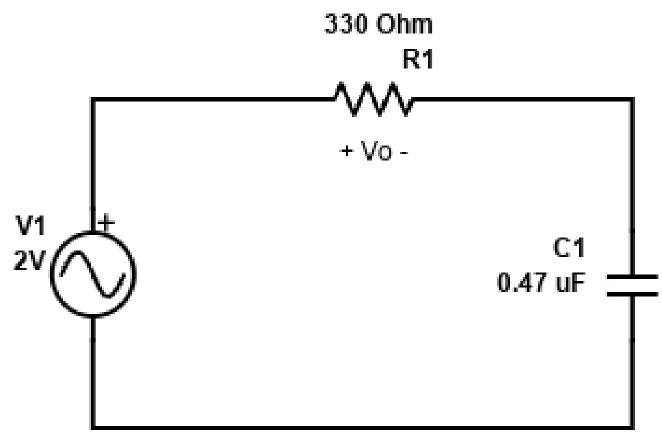
By sweeping the input frequency and measuring the output, we can observe the filter's gain vs frequency behavior and verify f_c experimentally.

Compilation of result and Calculations (with ckt diagram and verification of result):

Circuit Diagram:



Low Pass



High Pass

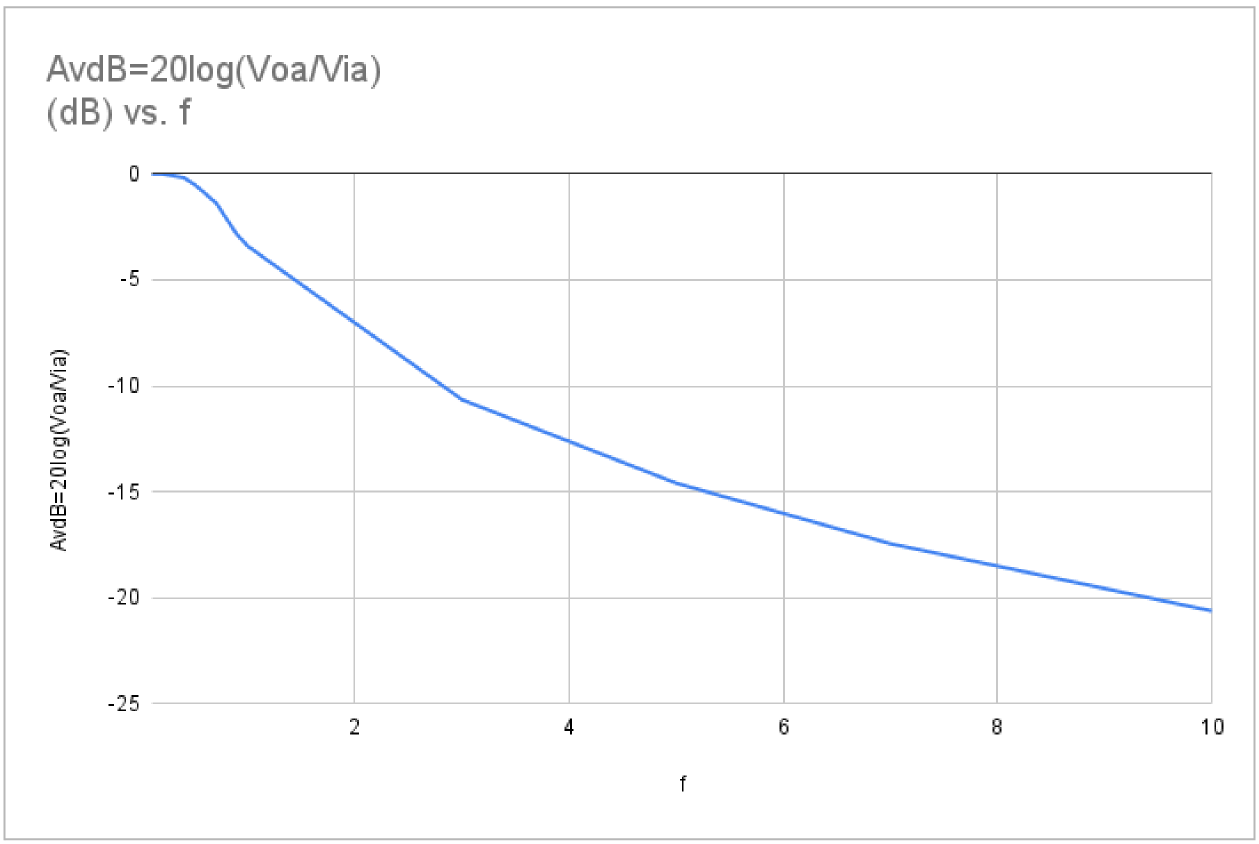
Data Table:

Low pass Filter

Table for theoretical cutoff frequency		
R (Ω) (using multimeter)	C (μF) (using multimeter)	$f_{ct}=1/2\pi RC$ (KHz)
330	0.46	1048.45

Table for gain			
f (kHz)	V_{ia}(V) (Peak value from oscilloscope)	V_{oa}(V) (Peak value from oscilloscope)	$A_{vdB}=20\log(V_{oa}/V_{ia})$ (dB)
0.1	2	2	0
0.2	2	2	0
0.4	2	1.96	-0.1755
0.5	2	1.89	-0.4914
0.7	2	1.71	-1.3607
0.9	2	1.435	-2.884
1	2	1.35	-3.414
3	2	0.586	-10.663

5	2	0.372	-14.60
7	2	0.268	-17.458
9	2	0.21	-19.576
10	2	0.186	-20.63



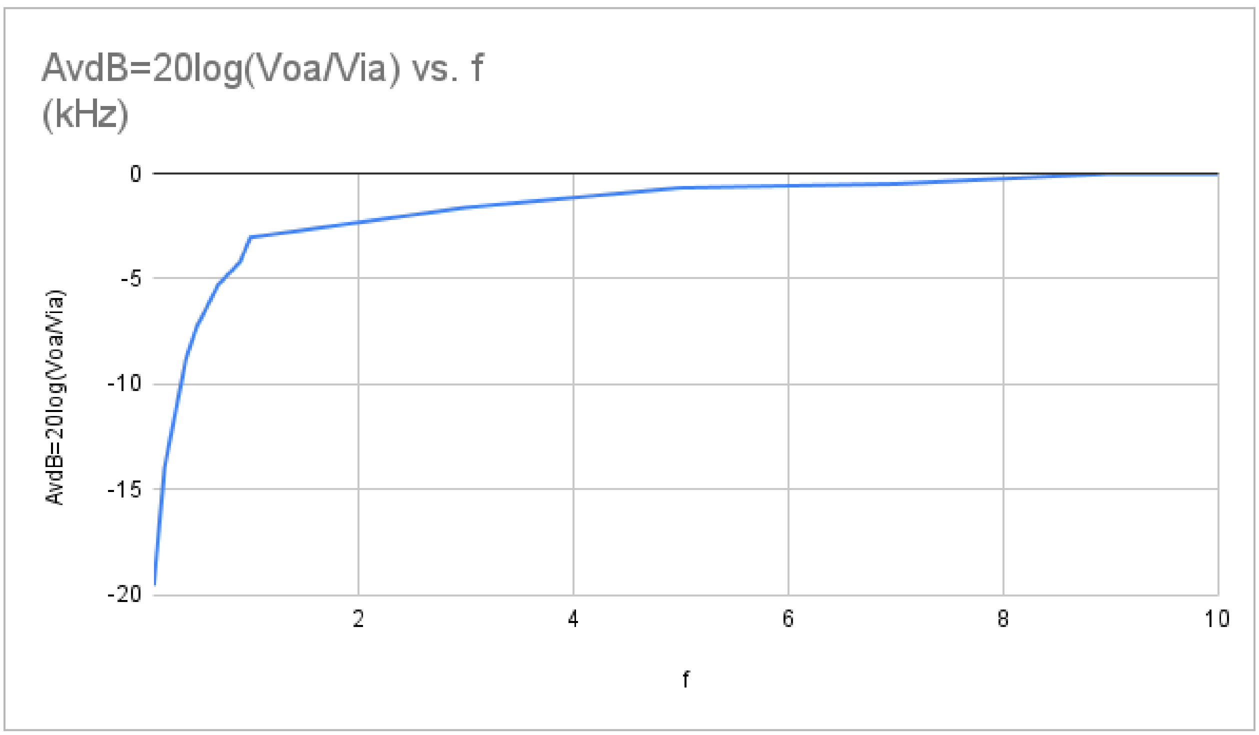
Graph : RC Low Pass Cut-off Frequency

High pass Filter

Table for theoretical cutoff frequency		
R (Ω) (using multimeter)	C (μF) (using multimeter)	$f_{ct}=1/2\pi RC$ (kHz)
330	0.46	1048.45

Table for gain			
f (kHz)	V_{ia}(V) (Peak value from oscilloscope)	V_{oa}(V) (Peak value from oscilloscope)	$AvdB=20\log(V_{oa}/V_{ia})$
0.1	2	0.21	-19.576
0.2	2	0.4	-13.979
0.4	2	0.726	-8.802
0.5	2	0.865	-7.280
0.7	2	1.090	-5.272

0.9	2	1.230	-4.223
1	2	1.41	-3.036
3	2	1.660	-1.618
5	2	1.85	-0.677
7	2	1.89	-0.491
9	2	2	0
10	2	2	0



Graph : RC High Pass Cut-off Frequency

Classwork Proof :

For Low pass :

<table>
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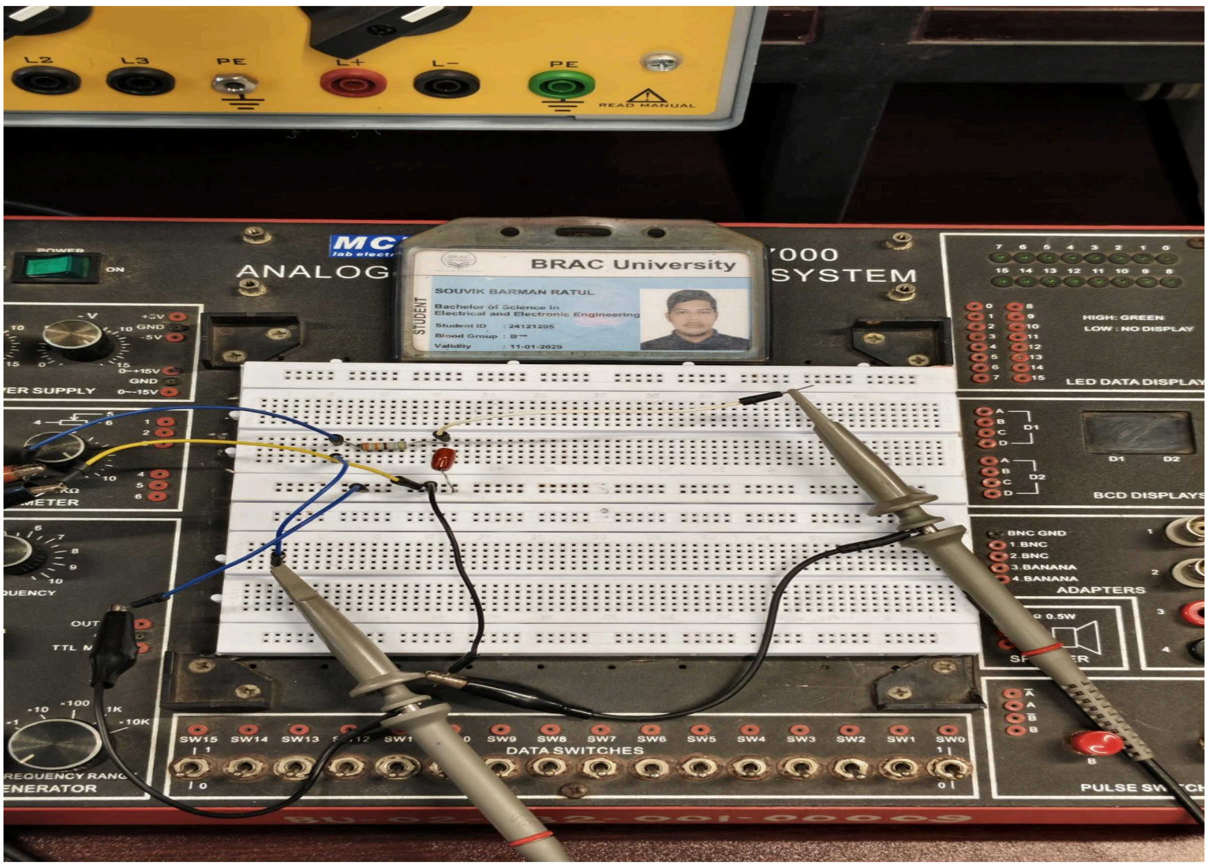
<tr><td colspan="3">
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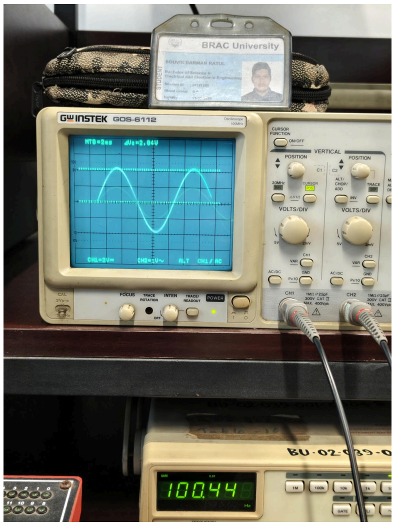
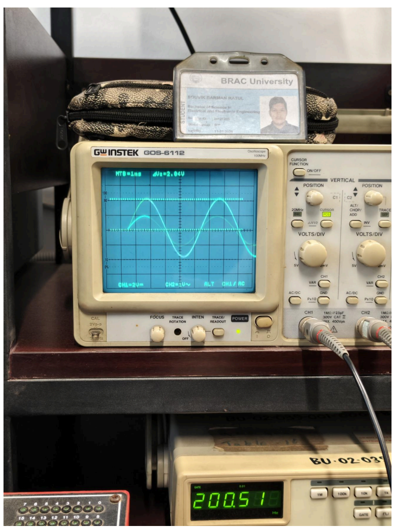
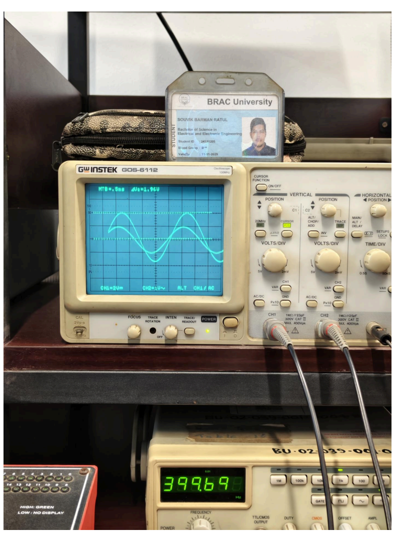
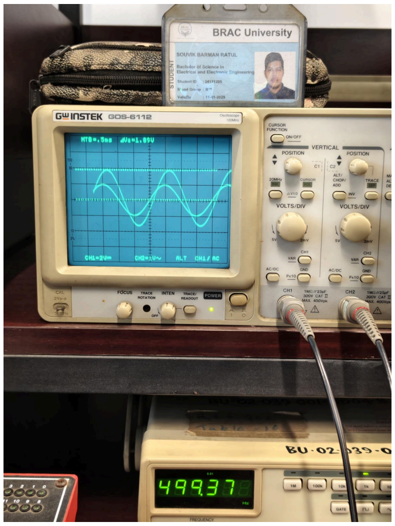
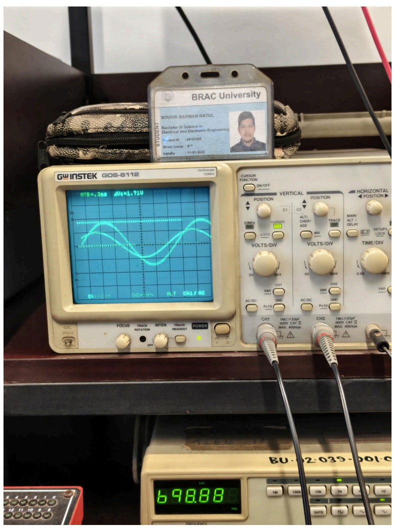
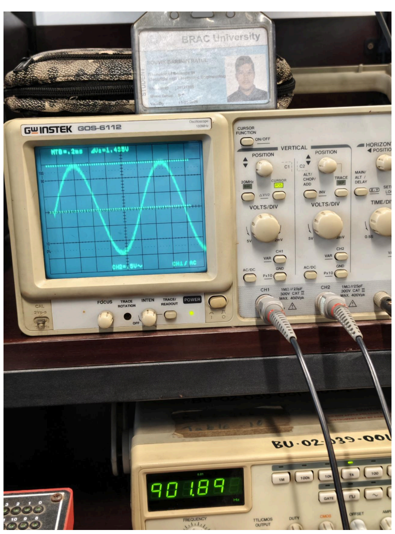
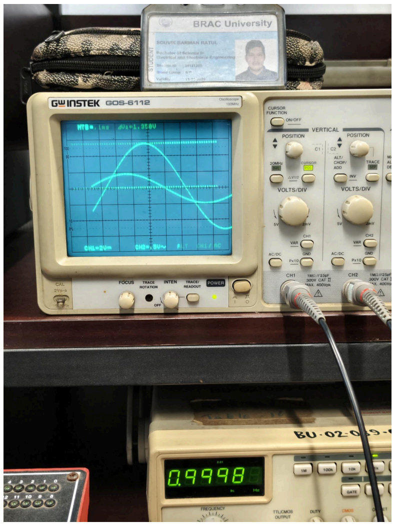
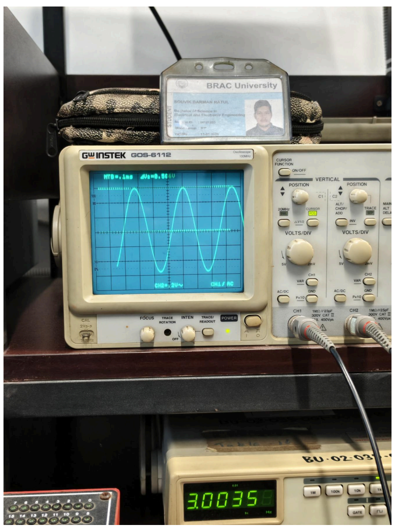
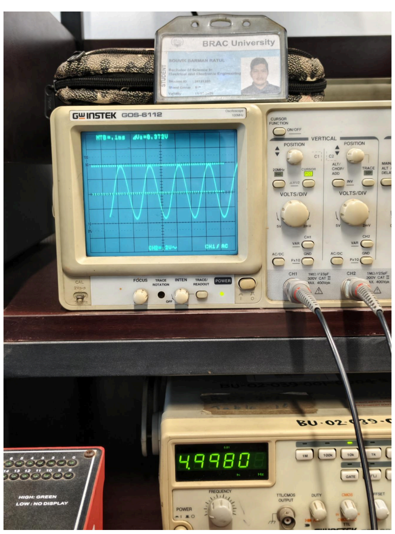
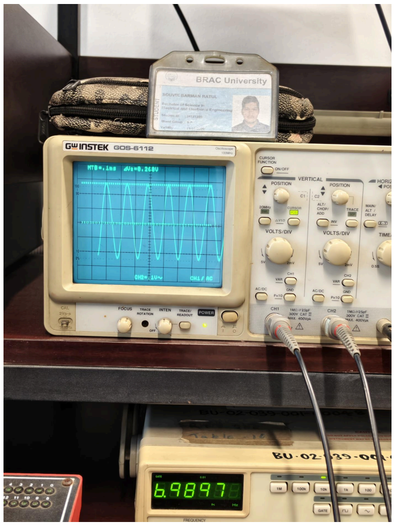
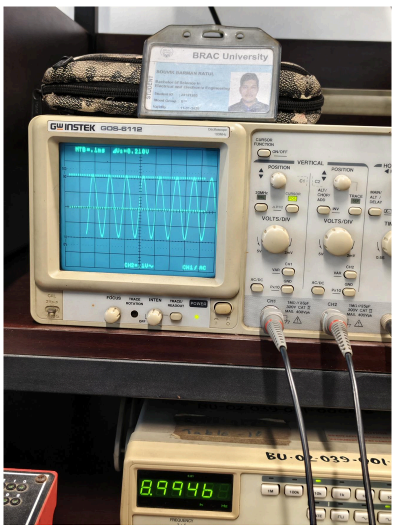
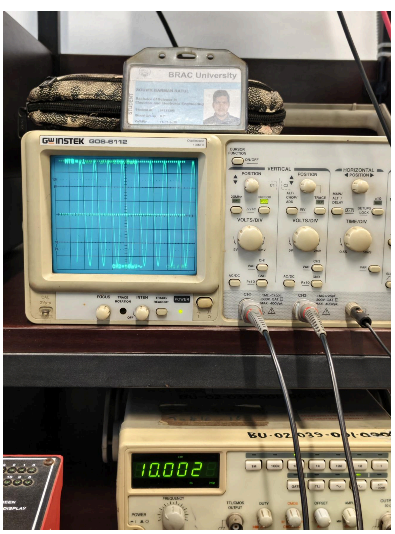
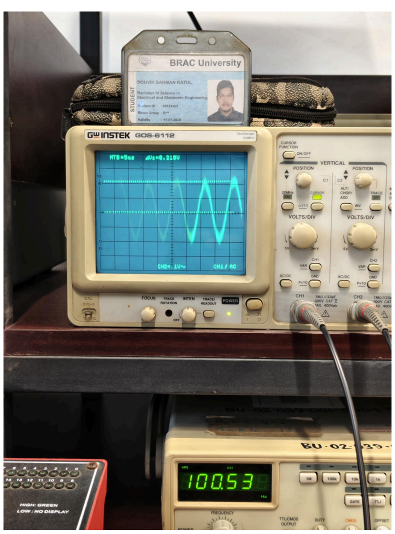
	0.1	2
	0.2	2
	0.4	1.96

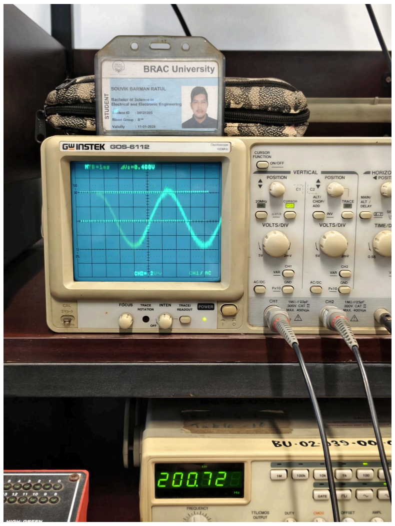
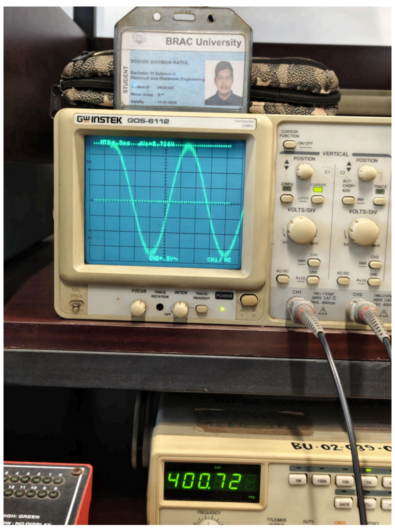
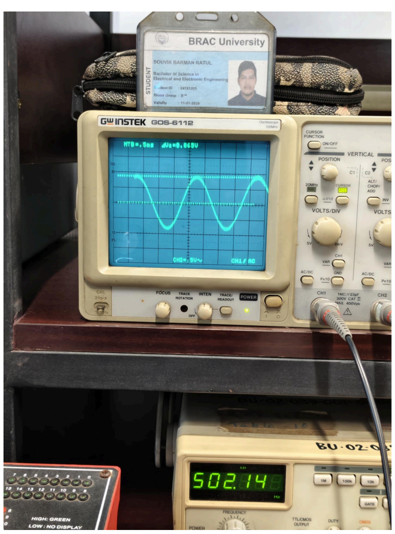
	0.5	1.89
	0.7	1.71
	0.9	1.435

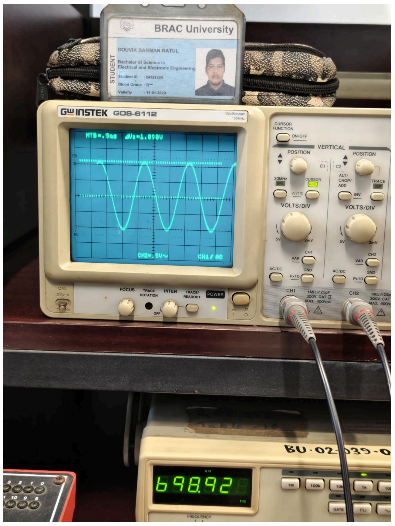
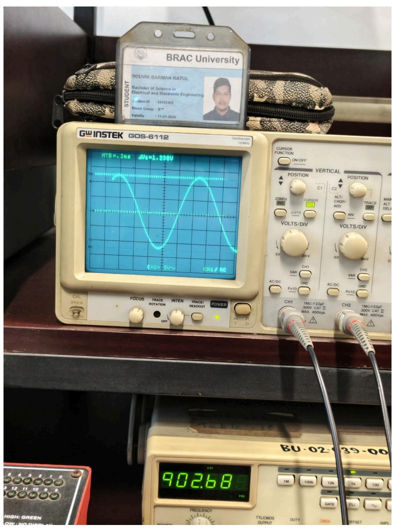
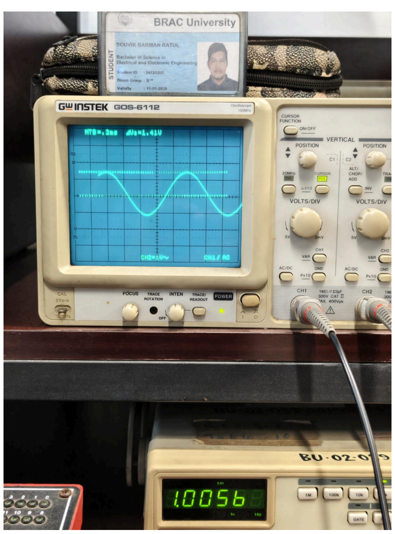
	1	1.35
	3	0.586
	5	0.372

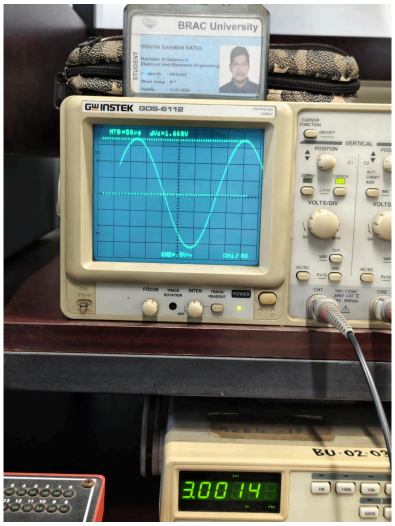
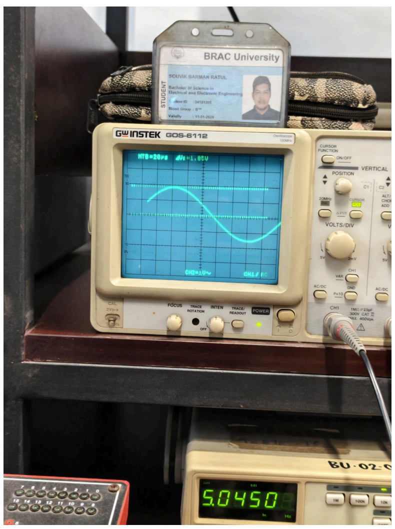
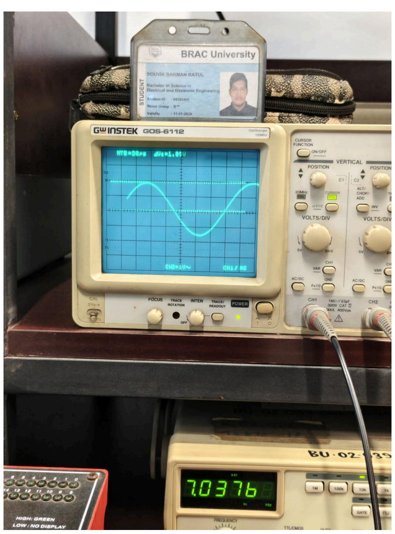
	7	0.268
	9	0.21
	10	0.186

For High pass :

Image Captured	Freq. we set (kHz)	Output/Peak (V)
		
	0.1	0.21

	0.2	0.4
	0.4	0.726
	0.5	0.865

	0.7	1.090
	0.9	1.230
	1	1.41

	3	1.660
	5	1.85
	7	1.89

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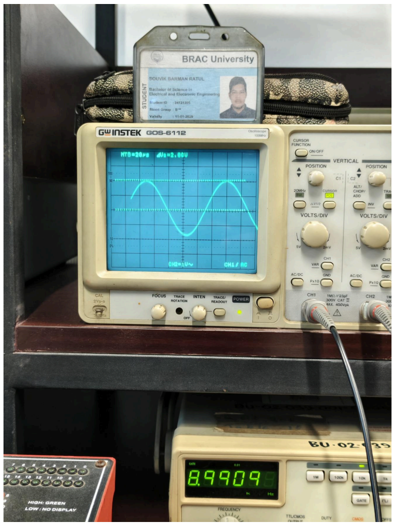
<tr><td>
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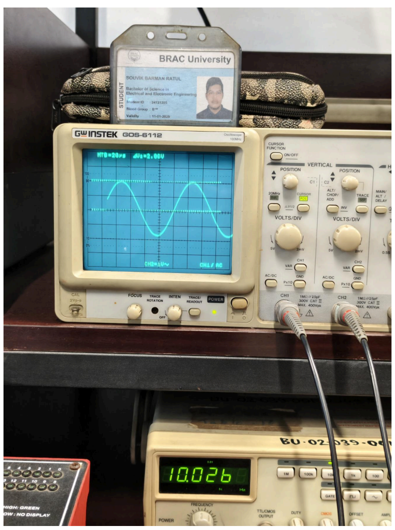
<tr><td>
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Appendix:

Brac University
Department of Electrical & Electronic Engineering (EEE)
EEE203L – Electrical Circuits II Laboratory

Experiment 5

Name of the Experiment

Passive Filters: Low Pass and High Pass Filters

Objective:

The purpose of this lab is to introduce students to Low Pass and High Pass Filters.

Apparatus:

- Resistance
- Capacitor
- Oscilloscope
- Oscilloscope Probe
- Signal Generator
- Signal Generator Probe
- Multi-meter
- Bread Board
- Wires for connection

Introduction:

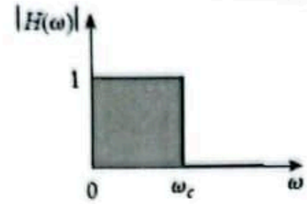
A filter is a frequency-selective circuit. Filters are designed to pass some frequencies and reject others.

Filter circuits depend on the fact that the impedance of capacitors and inductors is a function of frequency. There are numerous ways to construct filters, but there are two broad categories of filters:

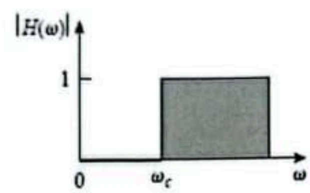
- Passive filters are composed of only passive components (resistors, capacitors, inductors) and do not provide amplification.
- Active filters typically employ RC networks and amplifiers (Op-Amps) with feedback and offer a number of advantages.

There are four basic kinds of filters:

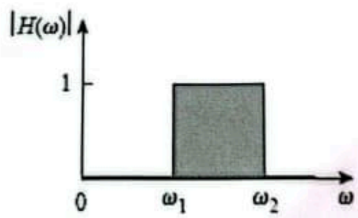
- Low-pass filter – Passes frequencies below a critical frequency, called the cutoff frequency, and attenuates those above.



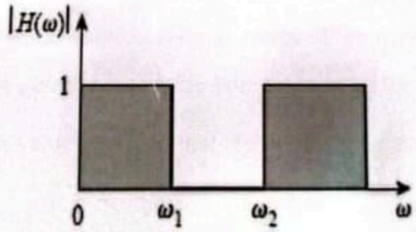
- High-pass filter - Passes frequencies above the critical frequency but rejects those below.



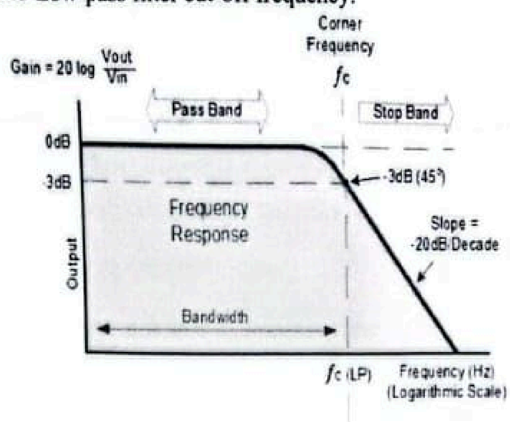
- Band-pass filter - Passes only frequencies in a narrow range between upper and lower cutoff frequencies.



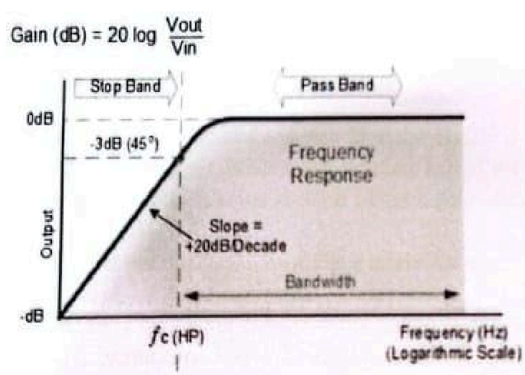
- Band-reject filter - Rejects or stops frequencies in a narrow range but passes others.



RC Low-pass filter cut-off frequency:



RC High-pass filter cut-off frequency:



- The cutoff frequency is the frequency at which capacitive reactance and resistance are equal (R = Xc), therefore fc = $1/2\pi RC$

- When our frequency response curve is given in terms of 'voltage vs. frequency':

  ''Cutoff frequency is the point at which the voltage level of the signal falls '' $\frac{1}{\sqrt{2}}$ or 0.707''

  times from its maximum value provided that the input signal amplitude remains same for all frequencies.''

- When our frequency response curve is given in terms of 'Signal power vs. frequency': ''Cutoff frequency is the point at which the power level of the signal falls '' $\frac{1}{2}$ or 0.5'' times from its maximum value.''

- When our frequency response curve is given in terms of 'gain vs. frequency': ''Cutoff frequency is the point at which the gain of the signal falls '' $\frac{1}{\sqrt{2}}$ or .707'' times from its maximum value.''

- When our frequency response curve is given in terms of 'gain (in dB) vs. frequency': ''Cutoff frequency is the point at which the gain (dB) of the signal falls ''-3 dB'' from its maximum value.''

Procedure:

1. Measure the values of the resistor and capacitor. Construct the circuit by connecting the capacitor and the resistor in series. Connect the oscilloscope to measure the voltage across the resistor.

2. Calculate the cutoff frequency from $fc= \omega c/2\pi =1/2\pi RC$. This should be the frequency at which the output power is equal to one half of the input power, or, equivalently, the output voltage is equal to 0.707 of the input voltage. Record this number.

3. Set the input voltage, the voltage across the signal generator, to 2 volts zero to peak.

4. Measure the output voltage of the circuit over a frequency range of 1 kHz to 10 kHz in 1 kHz increments. You will need to adjust the input voltage back to 2 volts as you change the frequency. Plot gain vs. frequency.

Low-pass Filter

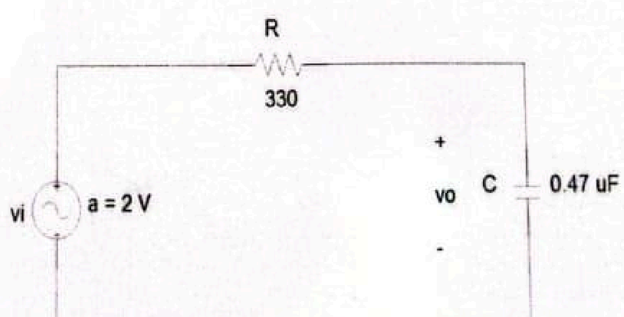


Table for theoretical cutoff frequency		
R (Ω) (using multimeter)	C (μF) (using multimeter)	$f_{ct} = \frac{1}{2\pi RC}$ (kHz)
330	0.46	1048.45

Table for practical cutoff frequency					
f (kHz)	V_{ia} (V) (Peak value from Oscilloscope)	V_{oa} (V) (Peak value from Oscilloscope)	$0.707V_{oa}$ (V)	V_{ia} (V) (Peak value from Oscilloscope)	f_{cp} (kHz)
0.1	2	2	1.414	1.41	1.023

Table for gain			
f (kHz)	V_{ia} (V) (Peak value from Oscilloscope)	V_{oa} (V) (Peak value from Oscilloscope)	$A_{vdB} = 20\log\frac{V_{oa}}{V_{ia}}$ (dB)
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0.5	2	1.89	-0.4914
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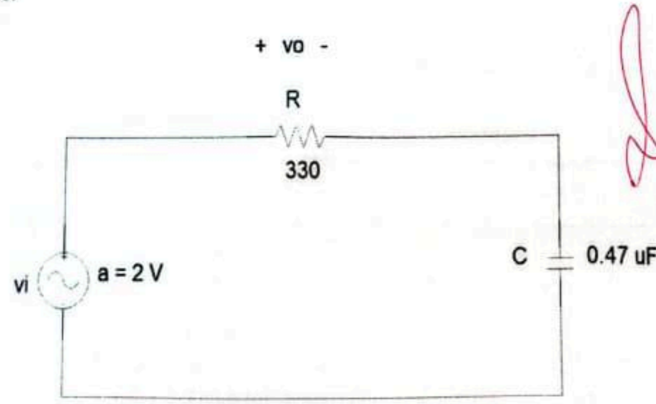
High-pass Filter



Saad
10/08/25

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3	2	1.660	-1.618
5	2	1.85	-0.677
7	2	1.89	-0.491
9	2	2	0
10	2	2	0

Questions:

1. What would happen to the value of fc if the value of the capacitor C for the low-pass and high-pass filters is increased?

2. What would happen to the value of fc if the input voltage is increased?

3. What would happen if you replaced the capacitors in the above circuits with inductors?

Saad 10/08/21

Conclusion:

The two circuits we built were a Low Pass Filter (LPF) and a High Pass Filter (HPF), using resistors and capacitors. We sent AC signals of different frequencies through them and measured the output voltage. For each filter, we found the frequency where the output dropped to about 0.707 (or $\frac{1}{\sqrt{2}}$) of its maximum value and compared this with the theoretical cutoff frequency. We found that:

1. The measured cut-off frequencies were close to the calculated values.

2. Any differences were likely due to component tolerances or measurement errors.
   This shows that passive RC filters work as expected: LPFs block high frequencies and HPFs block low frequencies, and both match the theory when phase and amplitude changes are considered.